All 11 papers: connected to your research

Your focus: preparing student teachers to combine subject knowledge, foreign-language skills, and teaching methods.

Based on the 11 PDFs in Papers.zip and your stated topic. No manuscript draft was supplied. Numbers match the archive order. The proposed applications are a synthesis to investigate, not a programme already proven by all 11 papers.

Paper / focus	Problem and contribution	Connection to your paper
1. Banegas, Montgomery & Raud (2025) Language Triptych	**Problem:** Student teachers need a practical way to connect subject learning with language support. **Contribution:** A qualitative study of 32 student teachers found the Triptych useful for organising lessons and connecting theory with practice.	Build a planning tool around language OF learning (subject language), FOR learning (interaction), and THROUGH learning (needs emerging during a lesson). **Evidence:** One English-teacher education programme in Argentina; adapt for subject majors.
2. Banegas & Arellano (2024) Teacher language awareness	**Problem:** Understanding language theory does not automatically lead to language-aware teaching. **Contribution:** Across three countries, 58 student teachers and five educators informed a model linking exploration, inductive input, reflection, and informed use.	Let students analyse a subject explanation, discover language patterns, reflect, and then design their own teaching activity. **Evidence:** A model derived from cases; its proposed cycle still needs evaluation.
3. Baecher, Farnsworth & Ediger (2014) Clear language objectives	**Problem:** Teacher candidates can struggle to state exactly what language pupils need for a subject lesson. **Contribution:** Analysis of 107 lesson plans found more difficulty with language objectives than content objectives; broad skills and vocabulary dominated.	Teach students to write a specific language objective alongside each subject objective, then check whether activities address both. **Evidence:** MA TESOL candidates in an ESL context; lesson plans do not establish classroom performance.
4. Lai & Wang (2025) Subject-specific workshops	**Problem:** Student teachers need practice combining science teaching with language support. **Contribution:** Fifteen pre-service teachers improved on assessed microteaching and written tasks after a six-week workshop combining pedagogical content knowledge and the Triptych.	Adapt its sequence: model a subject lesson, practise explanations and scaffolding, teach a short lesson, and receive feedback. **Evidence:** Small before/after study; teaching was to experts online, not a school class. See the reporting note on page 2.
5. Scherzinger & Brahm (2023) Competence goals	**Problem:** It is unclear which combined abilities bilingual subject teachers need. **Contribution:** A review of 79 reports identifies 16 competencies and develops a professional competence model.	Choose and justify your programme outcomes: accurate subject explanations, language support, teaching decisions, collaboration, and inclusive practice. **Evidence:** Broad secondary bilingual education review; it does not validate your assessment rubric.
6. He & Lin (2018) Collaborative mentoring	**Problem:** A subject teacher needs support to become attentive to language in teaching. **Contribution:** A science-teacher case links language-awareness development with repeated collaboration and dialogue with a teacher educator.	Organise joint planning and feedback. Subject and language educators discuss the same lesson with the student teacher. **Evidence:** An existing teacher in Hong Kong; adapt the model for pre-service students.

The remaining papers and their roles

Paper / focus	Problem and contribution	Connection to your paper
7. Kim & Graham (2022) Needs-based preparation	**Problem:** Training may overlook difficulties teachers report in integrating content and language. **Contribution:** A review of 43 articles identifies uneven attention to seven areas of competence and gaps between reported needs and professional development.	Start with a needs analysis. Check language, pedagogy, subject knowledge, organisation, relationships, collaboration, and reflection. **Evidence:** Teachers broadly, with much self-reported evidence; confirm local needs through actual teaching tasks.
8. Morton (2024) Conceptual foundation	**Problem:** Teacher education needs a clear account of how language awareness supports content learning. **Contribution:** This editorial synthesises six studies and highlights theory-practice connections, collaboration, beliefs, disciplinary language, and multilingual resources.	Explain why your model connects language knowledge with teaching decisions and attention to pupils' language backgrounds. **Evidence:** A conceptual editorial, not a new intervention trial. It discusses paper 2, so these are not independent evidence.
9. Wu, Chun & Lan (2026) Practice, feedback and revision	**Problem:** Being comfortable with technology does not ensure effective integrated teaching with it. **Contribution:** Forty-three pre-service teachers completed an 18-week CATERR programme with three cycles of practice, feedback, reflection, and refinement.	Adapt the repeated development cycle and combine self-assessment with observed teaching. Use digital tools where they help the lesson. **Evidence:** Technology-focused, mostly English majors, and no control group; evidence is not proof of your whole model.
10. Brzosko-Barratt (2019) Programme coherence	**Problem:** Separate subject, language, and teaching courses make integration difficult. **Contribution:** A five-year programme-design case examines shared CLIL units, school practice, educator cooperation, and varied student needs.	Make one lesson project connect coursework, subject and language feedback, and teaching practice. Offer support according to starting needs. **Evidence:** Early-primary preparation in Poland; educator coordination was itself a challenge.
11. Escobar Urmeneta (2013) Reflection on real interaction	**Problem:** A planned lesson may fail when pupils struggle and the student teacher cannot adapt. **Contribution:** A student-teacher case traces improved interaction through teaching, selecting and analysing recordings, and reflection.	Ask students to identify one moment of pupil confusion, explain what happened, revise their response, and try it in another lesson. **Evidence:** One student teacher; improvement depends on how strategies fit the classroom context.

An important reading note

In Lai & Wang (paper 4), Table 4 reports $p = .027$ and $p = .016$ as significant, although its note specifies $p < .003125$. These two results do not meet that stated threshold. The reflection and social-interaction survey changes were also not significant. Use the workshop design and observed improvements cautiously; avoid claiming that every outcome improved. Source: uploaded PDF page 8.

One coherent proposal for your paper

Working title: Developing Pre-Service Teachers' Integrated Subject and Foreign-Language Teaching Competence through Guided Planning, Practice, and Reflection.

The central argument

Student teachers need opportunities to connect what they know about a subject with the language pupils need to understand and discuss it. Your programme can make that connection explicit, then help students practise, receive feedback, and improve.

How all 11 papers contribute

Part of your proposal	What you would do	Papers
Identify needs	Use a short lesson task and interview to identify difficulties.	3, 5, 7
Develop language awareness	Analyse subject explanations and pupils' language needs.	1, 2, 8
Plan and model lessons	Connect subject goals, language goals, activities, and assessment; demonstrate examples.	1, 3, 4, 10
Practise with support	Teach, discuss the same lesson with mentors, and revise.	4, 6, 9, 11
Connect the programme	Use shared assignments and coordinated mentoring across courses.	6, 10

The training cycle

Diagnose needs > analyse content and language > plan > teach > receive feedback > reflect and revise > apply to a new topic.

A simple example: teaching evaporation

Subject goal: pupils explain that liquid water becomes water vapour.
Language OF learning: terms and explanations about evaporation.
Language FOR learning: expressions for comparing observations or asking for clarification.
Language THROUGH learning: new language needs noticed as pupils discuss an unexpected question. Record these and respond during or after the activity.

How to investigate it

Question: How does this programme support student teachers' integrated teaching competence, and which activities help them most?
Evidence: comparable teaching tasks before and after training, lesson-plan revisions, mentor feedback, and interviews. Include a new-topic task or later school lesson to examine transfer. Assess subject accuracy, explanation clarity, language support, and responses to pupil difficulties.

Your potential contribution is a carefully adapted and evaluated model for your population and subject. These papers justify its components; they do not by themselves establish that your combination is novel or effective. A comparison group would strengthen causal claims.

References and source locations

All 11 entries correspond to files in Papers.zip. PDF page numbers below refer to the uploaded files, counting cover sheets, rather than printed journal page numbers. Links open the publication records.

1. Banegas, D. L., Montgomery, D. P., & Raud, N. (2025). Student-teachers' understanding of language teaching through the CLIL Language Triptych. *Learning and Instruction, 95*, 102044. [Paper]
Reviewed source locations: PDF pp. 2-3, 5-9.

2. Banegas, D. L., & Arellano, R. (2024). Teacher language awareness in CLIL teacher education in Argentina, Colombia, and Ecuador: A multiple case study. *Language Awareness, 33*(4), 707-730. [Paper]
Reviewed source locations: PDF pp. 6-9, 18-21.

3. Baecher, L., Farnsworth, T., & Ediger, A. (2014). The challenges of planning language objectives in content-based ESL instruction. *Language Teaching Research, 18*(1), 118-136. [Paper]
Reviewed source locations: PDF pp. 7, 14-16.

4. Lai, C.-J., & Wang, Y.-F. (2025). Transformative impact of pre-service CLIL science workshops on pedagogical content knowledge and language teaching competencies. *Humanities and Social Sciences Communications, 12*, 1227. [Paper]
Reviewed source locations: PDF pp. 5-10.

5. Scherzinger, L., & Brahm, T. (2023). A systematic review of bilingual education teachers' competences. *Educational Research Review, 39*, 100531. [Paper]
Reviewed source locations: PDF pp. 1-4, 16-18.

6. He, P., & Lin, A. M. Y. (2018). Becoming a "language-aware" content teacher: Content and language integrated learning (CLIL) teacher education as a collaborative, dynamic, dialogic process. *Journal of Immersion and Content-Based Language Education, 6*(2), 162-188. [Paper]
Reviewed source locations: PDF pp. 1-2, 24-30.

7. Kim, H., & Graham, K. M. (2022). CLIL teachers' needs and professional development: A systematic review. *Latin American Journal of Content & Language Integrated Learning, 15*(1), e1515. [Paper]
Reviewed source locations: PDF pp. 2, 7-16.

8. Morton, T. (2024). Conceptualizing and developing teacher language awareness for content and language integration (TLA for CLI) across educational contexts. *Language Awareness, 33*(4), 673-683. [Paper]
Reviewed source locations: PDF pp. 2-5, 9-10.

9. Wu, Y.-j. A., Chun, D. M., & Lan, Y.-J. (2026). Cultivating CLIL preservice teachers' CALL competencies through a TPACK-based teacher training program. *ReCALL, 38*(1), 22-39. First published online in 2025. [Paper]
Reviewed source locations: PDF pp. 5-9, 15-16.

10. Brzosko-Barratt, K. (2019). Designing a pre-service CLIL teacher education program. *Kwartalnik Pedagogiczny, 64*(3), 173-193. [Paper]
Reviewed source locations: PDF pp. 8-14, 18-20.

11. Escobar Urmeneta, C. (2013). Learning to become a CLIL teacher: Teaching, reflection and professional development. *International Journal of Bilingual Education and Bilingualism, 16*(3), 334-353. [Paper]
Reviewed source locations: PDF pp. 5-6, 17-18.

Source note: paper 6 is an author manuscript with a title variant using "Teacher Professional Development". Its published citation uses "Teacher Education". Paper 8 is Morton (2024), not Morton (2016/2018) from the earlier reading list. Paper 9 appears in the 2026 journal issue and was first published online in 2025.